

ORIGINAL PAPER

Resource Allocation Techniques in Multibeam Satellites: Conventional Methods vs. AI/ML Approaches

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ABSTRACT

This paper presents a comprehensive survey of advancements in resource allocation techniques within the realm of multibeam satellites, focusing specifically on four key areas related to payload beam hopping, along with allocations of power, bandwidth, and beamwidth. It provides a comprehensive examination of traditional approaches alongside the innovative adoption of artificial intelligence and machine learning (AI/ML) methods to tackle these obstacles. A comprehensive analysis is carried out to investigate the possible approaches to enhance the resource allocation efficiency further. While acknowledging the plethora of topics within the multibeam satellite domain, this study deliberately narrows its focus to these four fundamental aspects, providing a nuanced understanding of the evolving landscape in satellite communications.

1 | Introduction

A single-beam satellite [1] uses a single large antenna having a large footprint to cover wide geographical area and caters to all users within the footprint. Monobeam satellite basic architecture is shown in Figure 1. Though efficient for broad coverage, single-beam satellites may suffer from congestion or limitations in capacity, as the entire footprint shares the same bandwidth (BW). Instead of relying on a single large antenna to cover large geographical area, multibeam satellites (MBS). MBS basic architecture is depicted in Figure 2. Kodheli et al. [2] employ multiple smaller antennas. In this scenario, the extensive coverage area is segmented into smaller regions, with

each beam directed towards a specific area, enabling individualized and accurate management of coverage and capacity distribution. This approach allows for better frequency reuse (FR) and can mitigate the congestion issues experienced with single-beam satellites. Spot beam satellites [3] take the concept of MBS further. Whereas the MBS offer broader coverage with multiple smaller beams, the spot beam satellites provide highly focused coverage to specific locations with intense, narrow beams and deliver even higher capacity and spectral efficiency compared to MBS. It allows for highly targeted coverage and optimal resource allocation (RA) within specific geographic areas. The spot beam satellites are mostly used in direct broadcast satellite television, broadband Internet

Abbreviations: A3C, asynchronous advantage actor-critic; ABC, adaptive beam control; ABIM, axiomatic-based interference model; AI, artificial intelligence; BA, bandwidth allocation; BFNA, beamforming network architecture; BH, beam hopping; BQP, binary quadratic programming; BW, bandwidth; BWA, beamwidth allocation; CBH, cluster-based beam hopping; CMADRL, cooperative multiagent deep reinforcement learning; CMOPSO, multiobjective particle swarm optimization algorithm with competition mechanism; CNN, convolutional neural network; CO, convex optimization; CSI, channel state information; DBF, digital beamforming; DCA, dynamic channel allocation; DL, deep learning; DLL, double-loop learning; DNN, deep neural network; DoCA, difference-of-convex approach; DPA, dynamic power allocation; DRA, dynamic resource allocation; DRL, deep reinforcement learning; DRL-DPA, deep reinforcement learning based dynamic power allocation algorithm; DRM, dynamic resource management; EE, energy efficient; FBMA, flexible beam management algorithm; FP, fractional programming; FR, frequency reuse; FRA, frequency resource allocation; GA, genetic algorithm; GEO, geostationary orbit; HTS, high-throughput satellites; L&O, learning and optimization; LCUD-BPD, low-complexity user density-based beam position design; LEO, low Earth orbit; MADRL, multiagent deep reinforcement learning; MARL, multiagent reinforcement learning; MBS, multibeam satellites; MDP, Markov decision process; MH, meta-heuristics; ML, machine learning; MODRL-TF, multiobjective DRL-based time-frequency resource allocation algorithm; NLP, nonlinear programming; NOFR, non-orthogonal frequency reuse; OFPA, optimized frequency plan algorithm; PA, power allocation; PBGA, priority-based greedy algorithm; PBVA, priority-based Viterbi algorithm; PPO, proximal policy optimization; PSO, particle swarm optimization; RA, resource allocation; SA, simulated annealing; SatCom, satellite communications; SBUM, smart beam-user mapping; SCA, successive convex approximation; SPGM, stochastic policy gradient method; SQP, sequential quadratic programming; UHTS, ultra-high-throughput satellites; VHTS, very high throughput satellites.

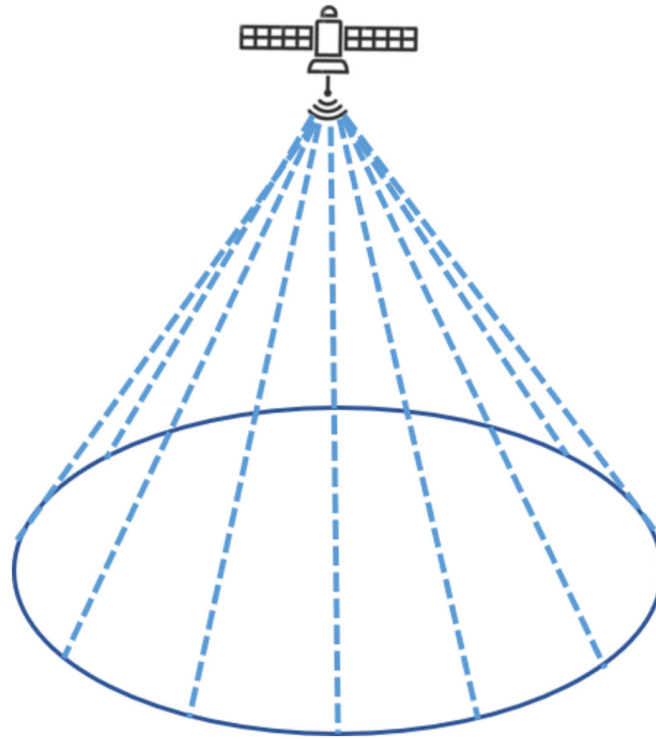


FIGURE 1 | Monobeam satellite system.

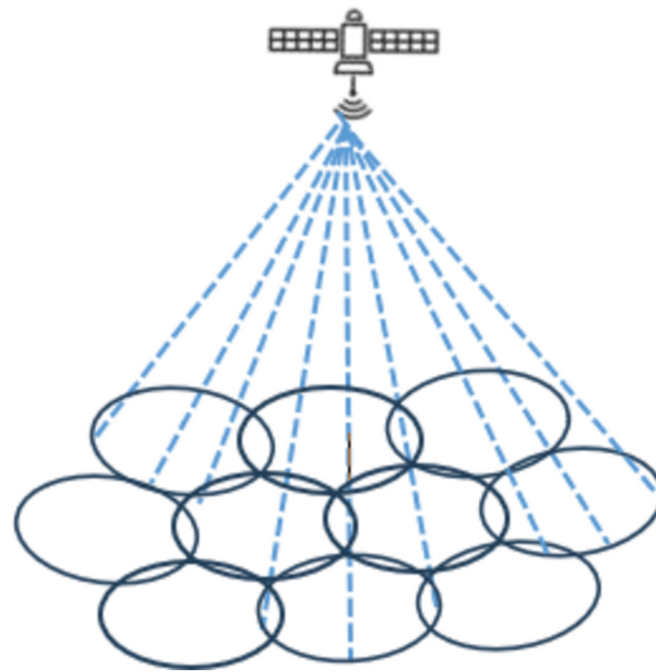


FIGURE 2 | Multibeam satellite system.

services, and other applications where targeted coverage and high capacity are essential. Both multibeam and spot beam satellites enhance satellite communication capabilities, and each technology has its advantages and is suited for different applications and deployment. The design of multibeam and spot beam satellites involves various complexities to ensure efficient and reliable operation and requires expertise in various disciplines, including antenna design, signal processing, power management, and system integration. Both require

sophisticated antenna systems capable of precisely shaping and directing multiple beams. These antennas must have high gain and beam-steering capabilities to focus the signal energy effectively. MBS requires advanced beamforming techniques to generate and steer multiple beams simultaneously. This involves complex signal processing algorithms to adjust phase and amplitude across the antenna array to achieve the desired beam shapes and directions. Spot beam satellites require even more precise beamforming and steering capabilities to

concentrate the signal energy into narrow, tightly controlled beams. This necessitates highly sophisticated beamforming algorithms and hardware. Both multibeam and spot beam satellites require careful frequency planning and spectrum management to ensure efficient use of available BW. FR techniques must be employed to minimize interference between adjacent beams and maximize spectral efficiency. Spot beam satellites particularly require meticulous coordination of frequency assignments to avoid interference and ensure optimal utilization of spectrum resources within each spot beam. Managing power distribution (PD) and allocation efficiently to ensure adequate signal strength across all beams while minimizing power consumption is essential for both multibeam and spot beam satellites. This involves sophisticated power management systems to regulate transmit power levels dynamically based on beam coverage and user demand. Robust control and management systems to monitor and adjust beam parameters, allocate resources, and maintain optimal satellite performance are also essential in these satellites. These systems must be highly reliable and capable of handling complex tasks autonomously. Several complexities inherent to the multibeam and spot beam satellite technologies dictate the need of artificial intelligence (AI) and machine learning (ML), which have the potential to greatly enhance the performance, efficiency, and capabilities of these satellites. Integrating AI and ML technologies into these satellite systems can offer significant performance improvements, operational efficiencies, and adaptive capabilities, ultimately enhancing the delivery of satellite communications (SatCom) services to users worldwide.

AI and ML algorithms can analyze real-time data on **user demand, traffic patterns, and atmospheric conditions** to dynamically adjust beam configurations and optimize beamforming parameters. This can **maximize spectral efficiency, minimize interference, and ensure optimal coverage for different regions and user groups**. These algorithms can also be used to intelligently allocate satellite resources, such as BW and power, among multiple beams or spot beams based on changing demand and priority levels. This dynamic resource allocation (DRA) can improve overall system performance and user experience. Besides, they have the capability to detect and mitigate interference from other satellites or terrestrial sources by analyzing received signal characteristics and adjusting beamforming parameters accordingly. This can enhance the resilience and reliability of SatCom systems in congested or hostile environments. Telemetry data and performance metrics from satellite subsystems can be very well analyzed to predict potential failures or anomalies before they occur. This proactive approach to maintenance can reduce downtime, extend satellite lifespan, and optimize operational efficiency. Autonomous operation such as beam steering, frequency planning, and power management of these satellites can be enabled, allowing them to adapt to changing conditions and optimize performance without human intervention to effectively maximize system efficiency. These techniques can optimize adaptive beamforming algorithms to adaptively adjust beam shapes and directions in real time based on environmental conditions and user requirements. This can improve link quality, coverage, and capacity in dynamic and challenging scenarios. Spectrum allocation and management of these satellite systems can be facilitated using these algorithms by

analyzing usage patterns, interference levels, and regulatory constraints. This can maximize spectrum utilization and minimize interference, enabling more efficient and reliable communication services.

There have been several surveys in the past linking AI/ML with SatCom. The authors [4] have highlighted the significance of AI within the realm of SatCom, emphasizing its capacity to mitigate challenges associated with RA, network supervision, security enhancement, spectrum regulation, and energy optimization in satellite networks. They extensively investigate a range of applications, encompassing beam hopping, strategies against jamming, prediction of network traffic, channel modeling, analysis of telemetry data, identification of ionospheric scintillation, interference management, remote sensing, behavioral modeling, integration of space-air-ground systems, and energy consumption optimization. In a similar vein, in [5], a survey is conducted focusing on AI/ML targeting solutions to problems in space applications. Their focus centers on mission planning, space exploration endeavors, and Earth observation initiatives, offering a categorization of existing obstacles within these domains. The authors [6] conduct a separate investigation into the implementation of AI methods within integrated terrestrial satellite networks, specifically focusing on the realm of extensive satellite network communications. They delineate the distinct characteristics of vast satellite networks and examine the hurdles linked with integrating them into the present communication framework. The survey illuminates cutting-edge AI methodologies deployed across diverse communication link layers, spanning the prediction of dynamically changing radio channels, classification and sensing of spectrum, detection and demodulation of signals, optimization of intersatellite and satellite access networks, and bolstering network security. Furthermore, the authors outline future paradigms and provide insights into how these mechanisms can be practically implemented in networks. In [7], a comprehensive exploration of SatCom is presented, challenges encountered with conventional methods and their limitations are discussed, and subsequently AI/ML-based approaches are proposed. The authors delve into the merits and demerits of onboard and on-ground AI/ML-based architectures. They investigate cutting-edge hardware solutions aimed at integrating AI/ML into operational satellite systems. Moreover, they discuss the enduring advancements of AI/ML within the realm of SatCom and nonterrestrial networks while also outlining potential avenues for future research. This overview offers an intricate analysis of the prospects and obstacles in leveraging AI/ML to augment the efficacy of nonterrestrial networks. Mostly, the previous surveys have covered the holistic aspects of SatCom in general, concerning ML/AI. In recognition of these surveys, we have sought to refine our focus by centering our survey on the optimization of resources, such as BW, power, beamwidth, and spectral efficiency, specifically in multibeam and spot beam satellites.

The survey paper represents the outcome of a methodological literature review **focusing on optimization of resources in MBS, encompassing both conventional and AI/ML methods**. Section 2 encompasses the work conducted in MBS for the optimization of resources through conventional methods and outlines their limitations and shortcomings. Next, it delves into the utilization of AI/ML methodologies. In Section 3, the obstacles

and opportunities within the realm of MBS are discussed examining both conventional methods and the applications of AI/ML techniques.

2 | Techniques in Multibeam Satellites

In this section, framework for the proposed survey is presented with the help of Tables 1 and 2. In Table 1, the status of traditional and AI/ML techniques is presented pertaining to **system performance such as resource utilization, quality of service (QoS), and responsiveness**. In Table 2, under various domain resources, the traditional techniques and AI/ML techniques employed by several researchers are presented along with respective references. Further, each of the spectrum allocation techniques and power management technique is also elaborated separately.

2.1 | Adaptive Beam Hopping

Adaptive payload BH in MBS refers to a dynamic and flexible approach where the satellite's payload can intelligently switch or adapt its communication beams in response to changing user

demand, environmental conditions, or operational requirements. This adaptive capability enhances the satellite's efficiency by optimizing beam coverage and RA, ensuring effective and responsive communication services across diverse geographic areas. Innovative developments in high-throughput satellite (HTS) technology involve producing numerous small beams, strategically reallocating the satellite spectrum that is accessible across the range area to significantly improve overall satellite capacity. Table 3 exhibits the challenges in BH and aspects affected in MBS. To address significant adjacent-beam interference within HTS systems, a notable solution has emerged: segmenting the entire spectrum into distinct bands, or "colors," separated by frequency allocation and polarization, also known as frequency reuse (FR). In the emerging era of UHTS, it is anticipated that the existing coverage zone will be subdivided into a greater quantity of smaller beams. Decreasing the beam pattern angle focuses on smaller areas, resulting in increased capacity gains while maintaining lower capital expenditure (CAPEX) costs. Nevertheless, the division into smaller areas raises the likelihood of no demand within particular beam spots at certain times. In such instances, intelligently switching off the corresponding no-demand beams can reduce operating costs—a core principle in the implementation of BH technology for the next era of satellite systems.

TABLE 1 | System performance, traditional vs. AI/ML techniques.

System performance	Traditional techniques	AI/ML techniques
Resource utilization	A heuristic approach to find the Lagrange multiplier [8] Dual methods for convex optimization problem [9] Iterative algorithm on duality theory [10] An axiomatic-based interference model [11] Multiobjective optimization, to enhance efficiency via a Pareto front, and proposing a dual-stage optimization structure [12] A novel genetic algorithm with repair functions [13] A successive convex approximation (SCA) algorithm [14] Use of a parallel multibeam scheme with optimum power allocation [15] Dinkelbach method and SCA [16] A nonconvex mixed-integer nonlinear program via iterative quadratic transformation [17] The equivalent channel and water-filling algorithm multibeam multicast nonorthogonal multiple access (MB-MC-NOMA) scheme [18]	Machine learning algorithms for managing flexible payload [19] Neural network for automatic management of a flexible payload architecture [20] DRL-based dynamic power allocation (DRL-DPA) algorithm [21] DRL using the proximal policy optimization algorithm [22] Mixed-integer programming issue modeled as a stochastic game, utilizing a MARL approach [23] Markov decision process (MDP)-based DRL algorithm using asynchronous advantage actor-critic (A3C) [24] A stochastic policy gradient-based deep neural network (DNN) approach [25]
QoS	Energy efficiency (EE) optimization [26] A nonconvex optimization structure [27] Smart beam-user mapping using Dirichlet distribution [28] A modified version of the simulated annealing (SA) algorithm [29] A dual-layer precoder and cluster-based power allotment scheme [30]	Convolutional neural networks (CNN) [31]
Responsiveness		Comparison of AI-based dynamic algorithms with conventional algorithms [32] Combine conventional optimization and deep learning to reduce computational complexity [33]

TABLE 2 | Domain resources, traditional vs. AI/ML techniques.

Domain resources		Traditional techniques	AI/ML techniques
Spectrum allocation	Beam hopping (BH)	Time space flexibility [34] PBGA scheme [35] Load balancing algorithm [36] LCUD-BPD scheme [37] PBVA scheme [38] Cluster-based hopping [39] Binary quadratic programming [40] Window-based algorithm [41]	DL-based algorithm [42] DLL using deep neural network [43] CMADRL framework [44] Deep beam leveraging DRL [45] Learning and optimization approach [46]
	Bandwidth allocation	Meta-heuristics [12] NOFR vs. BH analysis [34] Frequency RA method [47] Smart beam-user mapping [28] FBMA [48] CO method [27] PSO algorithm [49] CMOPSO algorithm [50] Precoding with FR [51] Difference-of-convex approach [52] OFPA [53]	MDP and DRL [54] DeepCA approach [55] MODRL-TF [56] CMADRL [44] RL and DNN [57] Analytical optimization with DL [33]
	Beamwidth allocation	Beamforming network architecture [58] Contoured beam generation [59] Beamforming networks [60] Greedy graph coloring [61] Beam coverage [62] DBF [63] ABC [64] Trunking systems [65] Spot beam distance optimization [66]	CNN's dynamic beam traffic [31] RL and DRL exploration [67] Adaptive pattern plan ML [68]
Power management	Adaptive payload—power allocation	Fractional programming, mean energy efficiency [69] Convex optimization by dual methods [9] Axiomatic-based interference model [11] Use of genetic algorithm [13] Energy efficiency optimization [26] Convexifying the nonconvex optimization problem using the Dinkelbach method and sequential convex approximation [16] Nonconvex mixed-integer nonlinear program via iterative convex transformation [17] Intelligent beam-user mapping [28] Parallel multibeam scheme Simulated annealing algorithm [29] A dual-layer precoder and cluster-based power allotment scheme [30] NOMA and water-filling algorithm [18] Sequential quadratic programming-based alternating optimization algorithm [70]	Application of ML [19] Neural network [20] Convolutional neural network [31] DRL-based dynamic power allocation [21] Proximal policy optimization based on DRL [22] Combination of conventional optimization and deep learning [33] Multiagent reinforcement learning approach [23] Markov decision process based DRL algorithm using asynchronous advantage actor-critic [24] Stochastic policy gradient based deep neural network [25]

The authors [71] have conducted a comprehensive analysis of BH in low Earth orbit (LEOs; addressing challenges, merits, system components, and architecture). The authors [72] delve into the

advantages of BH, conducting an in-depth examination that encompasses a comparative evaluation of performance and capacity capabilities between a system employing BH and equivalent

systems without hopping, all aimed at providing heightened flexibility. BH plays a pivotal role in enabling SatCom to adapt to time-varying communication demands dynamically influenced by factors such as temporal factors, mobility status, and atmospheric conditions. Unlike the fixed lighting approach typical in

conventional multibeam satellite setups, the satellite dynamically lights up a specific set of beams based on a predetermined schedule. BH technology entails employing a reduced number of active beams aligned with the BH pattern design to effectively cater to ground terminals within the satellite's coverage zone. This approach optimally utilizes available satellite resources, providing services that cater to heterogeneous demands [71, 73]. The main challenge in BH pattern design stems from the need to explore a vast search space to identify the most optimal patterns. With the number of potential BH patterns escalating exponentially alongside the number of beams [74], the exploration becomes increasingly complex. In this scenario, ML emerges as a promising approach, providing an alternative to traditional develop algorithms to optimize resource management in complex wireless networks.

Traditional approaches to BH in MBS often involve the use of static or semistatic beam patterns and predefined schedules. These techniques rely on historical data and predicted demand patterns to allocate resources. Though effective in many scenarios, these methods can be limited in their ability to adapt quickly to real-time changes in user demand or environmental conditions. The optimization of BH patterns using conventional techniques can also be computationally intensive, given the vast search space of possible configurations. Figure 3 exhibits the traditional/conventional techniques for BH used in MBS in recent times.

In recent years, ML has emerged as a powerful tool to enhance the performance and efficiency of BH systems. ML techniques can dynamically analyze large datasets, including real-time and historical data, to predict demand patterns and optimize resource allocation. By leveraging ML, BH systems can adapt more rapidly to changing conditions, improving overall satellite capacity and service quality. ML models can also reduce the computational complexity involved in exploring the vast search space of BH patterns,

TABLE 3 | Challenges in BH and aspects affected.

Challenges	Aspect of beam hopping (BH) affected
Adjacent beam interference	Signal quality, spectrum efficiency
Beam coverage challenges	Coverage area, resource allocation, service continuity
Adapting to time-varying communication demands	Dynamic resource allocation, QoS maintenance
Vast search space for optimal patterns	Computational complexity, real-time adaptation
Resource management complexity	Power allocation, bandwidth allocation, system efficiency
Signal quality degradation	Communication reliability, user experience
Synchronization issues	Timing coordination, system reliability
Increased latency	Communication speed, user experience, system responsiveness

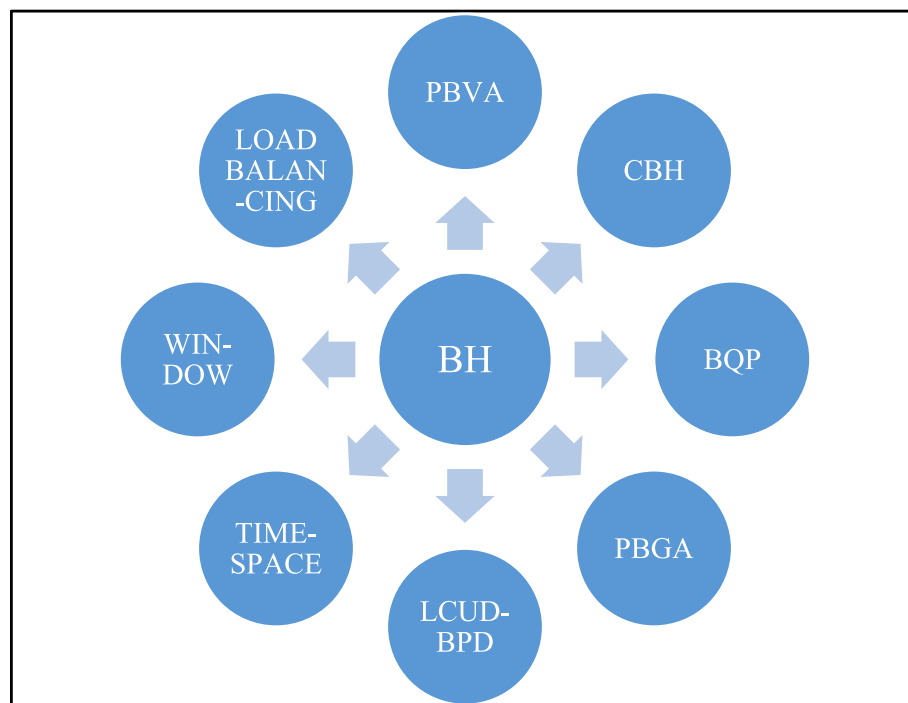


FIGURE 3 | Non-ML techniques in BH.

providing more efficient and effective solutions than traditional methods. Figure 4 exhibits the ML techniques for BH used in MBS in recent times. Table 4 exhibits the system performance of traditional techniques and ML/AI techniques used in BH.

It can be observed that in the field of BH in MBSs, most of the work has been carried out using traditional or conventional techniques, compared to the AI and ML techniques. This clearly indicates an opportunity to explore and apply AI and

ML methods to enhance system performance in BH. Chart 1 exhibits the resource utilization in BH using double-loop learning method [21].

2.2 | Adaptive Payload-PA

In the foreseeable future, advancements in satellite technology are anticipated to usher in a new era of “flexible payload-PA”

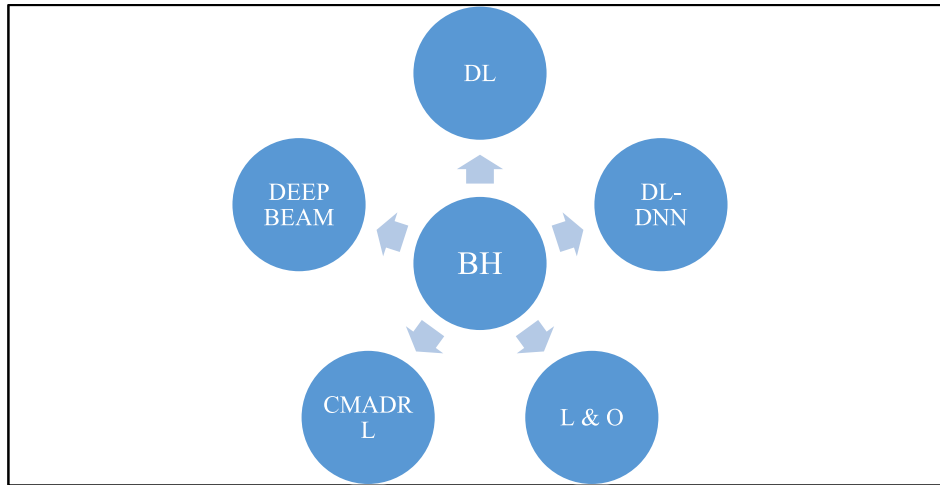


FIGURE 4 | ML-driven techniques in BH.

TABLE 4 | System performance of traditional and ML/AI techniques in BH.

System performance	Traditional techniques used	AI/ML techniques used
Resource utilization	Time space flexibility [34] Iterative minimal co-channel interference, iterative SINR maximization algorithms, and network coding based techniques at the link layer [75] Penalty function based on interference framed in BQP and two convexification frameworks SDP and MPMM alongside greedy algorithm [40] Window-based algorithm and the DNN-based approach [41] Greedy-based load balancing scheme [36] Priority-based Viterbi algorithm [38] Cluster-based BH approach [39]	Double-loop learning method [43] Collaborative multiagent deep reinforcement learning (MADRL) framework [44] DRL-based approach greedy algorithm [76] DRL-based algorithm, DeepBeam [45]
QoS	DWP-APM algorithm with roulette, stochastic exploration, and nonlinear weighting approaches [77] Combination of the hopping beam technology and the signal fence technology [78] Binary BH patterns using methods based on regular or irregular weights, Viterbi, and greedy algorithms [35]	DL-based optimization algorithm [42] A hybrid learning-and-optimization (L&O) technique [46]
Responsiveness	Dynamically division of beam position as p-center problem [79] Low-complexity user density-based beam position design (LCUD-BPD) scheme using an iterative user selection and smallest BP radius (USSBR) algorithm [37]	

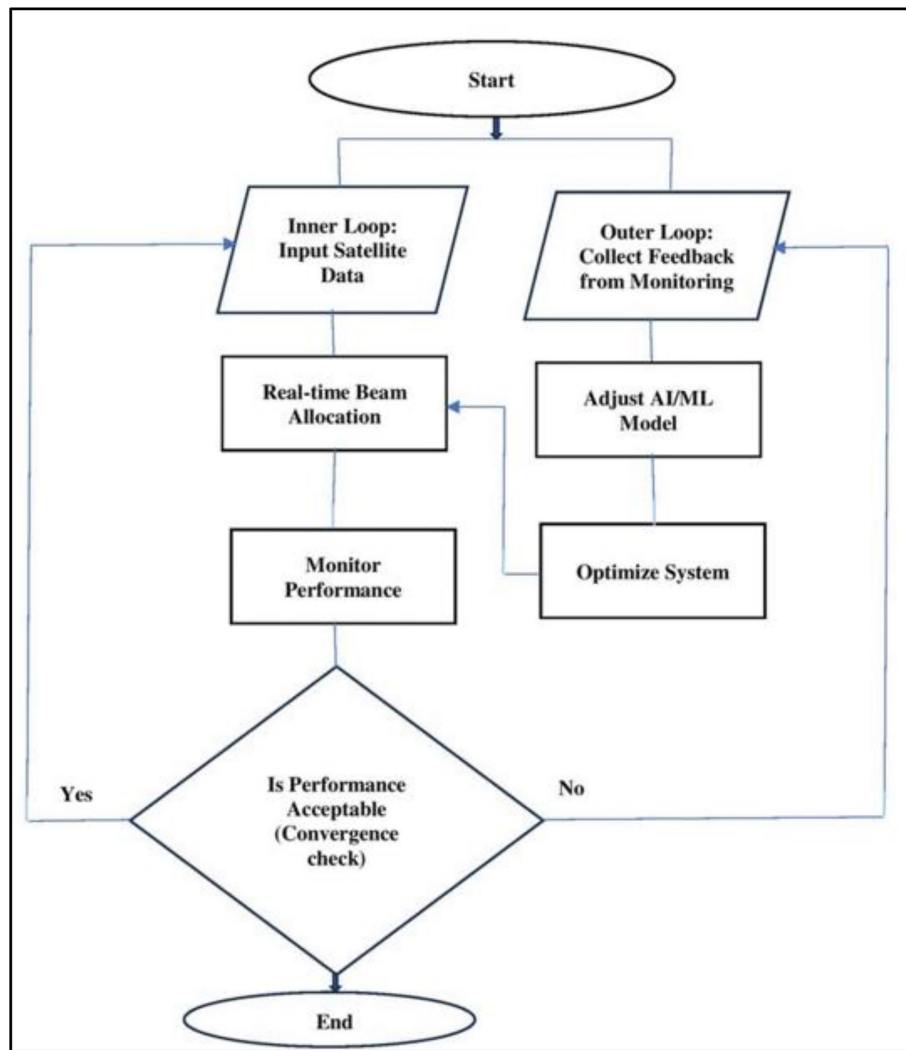


CHART 1 | Resource utilization using DLL method.

TABLE 5 | Challenges in BH and aspects affected.

Challenges	Aspect of power allocation (PA) affected
Power distribution efficiency	Optimal distribution of power across beams
Dynamic load balancing	Adjusting power allocation based on varying traffic
Signal interference management	Minimizing interference between beams
Real-time adaptation	Quick response to changing demands
Limited power resources	Efficient use of limited power
Complexity in resource allocation	Managing power allocation complexities
Beam coverage optimization	Ensuring effective coverage with allocated power
Maintaining service quality	Keeping consistent service quality across beams

for multibeam satellites. This innovative approach will empower these satellites to dynamically allocate and manage their payload and power resources, optimizing performance and adaptability to meet evolving communication and data demands. As a result, the satellite industry is poised to witness enhanced efficiency and responsiveness, fostering a more resilient and capable space infrastructure. Figure

exhibits potential issues in flexible payload power allocation (PA) in MBS. Table 5 exhibits the challenges and aspects of PA affected in MBS. An ML-based techniques for automating SatCom operations is presented in [74] to achieve notable results, including a 32% decrease in unmet capacity through flexible payload optimization, a 44% reduction in interference false detection, and a 10% improvement in prediction accuracy

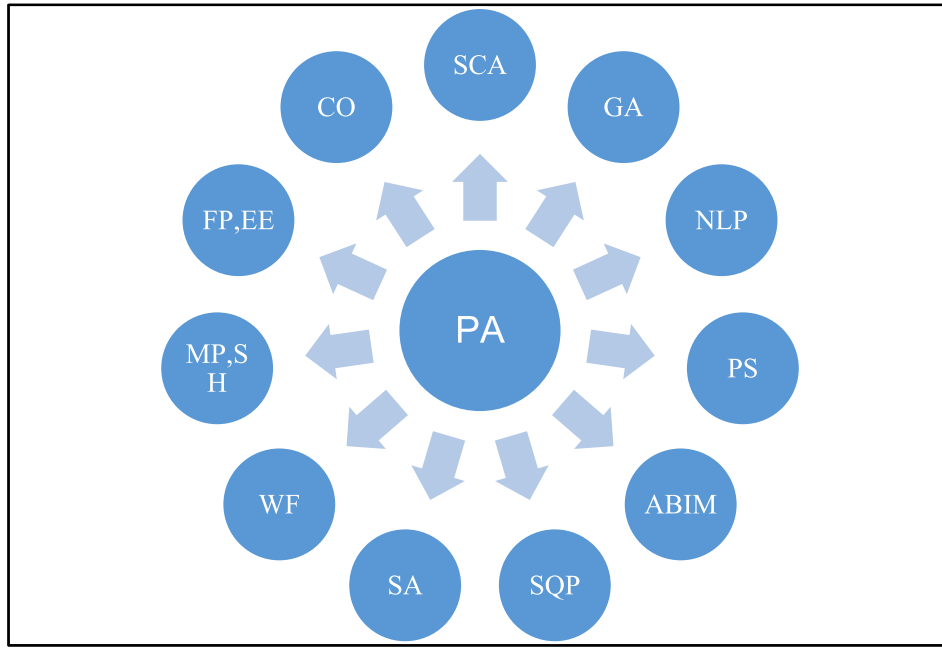


FIGURE 5 | Non-ML techniques for PA.

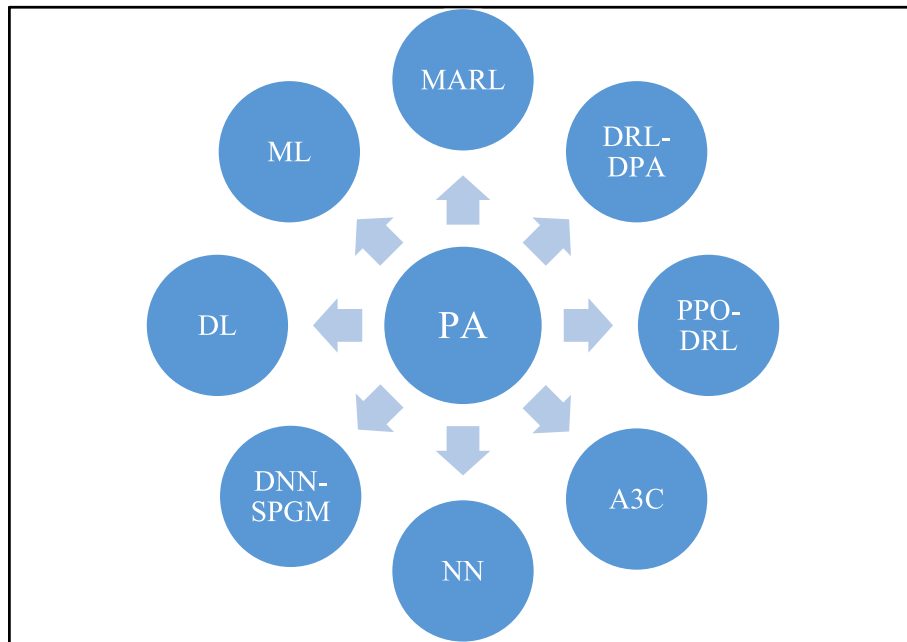


FIGURE 6 | ML-driven techniques for PA.

for traffic. These advancements showcase the potential of ML in enhancing efficiency and accuracy in SatCom systems. In [80], the necessity of adaptive approaches like flexible payload operations is emphasized based on the growing demand for BW. Addressing challenges in traditional geostationary missions, the shift towards flexible payloads, exemplified by initiatives like the quantum satellite, introduces dynamic transmission adjustments, necessitating automated payload management tools for optimal efficiency in addressing changing user demands and maximizing capacity. Highlighting the imperative need for flexible PA in very high throughput satellite (VHTS) systems, use of neural networks (NNs) to autonomously manage RA in dynamic satellite environments is

presented in [20]. As VHTS systems aim for 1 Tb/s capacity, flexible payloads become essential to optimize the satellite's full capacity over time, addressing nonuniform and evolving traffic demands through an intelligent NN-based switch for efficient and low-computational-cost RA. Recognizing the anticipated surge in traffic demand within VHTS systems, the authors [81] underscore the crucial need for flexible PA to address dynamic and nonuniform requirements over the service area. Emphasizing the autonomy and low-latency demands of resource management in flexible payload architectures, the paper advocates the application of supervised ML, employing a NN-based classification algorithm for efficient power and BW management.

Traditional techniques for adaptive power allocation in MBS include methods like equal power allocation, where power is evenly distributed among all beams, and static power allocation, where power levels are set based on anticipated traffic patterns but do not adapt to real-time changes. Another approach is using predefined power masks, where specific power levels are assigned to beams based on historical data or regulatory constraints. Although these methods are straightforward, they often fail to optimize power utilization effectively in dynamic environments. Figure 5 exhibits the traditional/conventional techniques used for adaptive power allocation in MBS.

AI and ML techniques in MBS systems for adaptive power allocation enable dynamic and intelligent distribution of power based on real-time data. These methods can predict traffic demand, optimize power allocation across beams, and adapt to changing channel conditions. Techniques like reinforcement learning and NNs are often used to continuously learn from the environment, making real-time decisions that improve efficiency, maximize coverage, and enhance overall system performance by responding to the dynamic nature of satellite communication networks. Figure 6 exhibits the AI/ML

techniques used for adaptive power allocation in MBS. Table 6 exhibits the system performance of traditional techniques, and AI/ML techniques in PA.

It can be observed that in the field of PA in MBS, a significant amount of work has still been done using traditional or conventional techniques. However, ML and AI methods have also been applied to improve system performance compared to traditional approaches. This presents an opportunity to further explore and apply ML and AI techniques to enhance performance in power allocation.

Chart 2 exhibits the resource utilization in PA using DRL-based dynamic power allocation (DRL-DPA) [21].

2.3 | Adaptive Payload-BA

As the need for higher data rates grows and the constraints of existing spectrum become more apparent, the distribution of BW has become one of the critical aspects in MBS. As the next generation of HTS is being designed to offer high data rates while controlling costs, the efficient allocation of flexible

TABLE 6 | System performance of traditional and ML/AI techniques in PA.

System performance	Traditional techniques used	AI/ML techniques used
Resource utilization	A heuristic approach to find the Lagrange multiplier [8] Dual methods for convex optimization problem [9] Iterative algorithm on duality theory [10] An axiomatic-based interference model [11] Multiobjective optimization, to enhance efficiency via a Pareto front, and proposing a dual-stage optimization structure [12] A novel genetic algorithm with repair functions [13] A successive convex approximation (SCA) algorithm [14] Parallel multibeam scheme with optimum power allocation [15] Dinkelbach method and SCA [16] A nonconvex mixed-integer nonlinear program via iterative quadratic transformation [17] Equivalent channel and water-filling algorithm multibeam multicast nonorthogonal multiple access (MB-MC-NOMA) scheme [18]	Machine learning algorithms for managing flexible payload [19] Neural network for automatic management of a flexible payload architecture [20] DRL-based dynamic power allocation (DRL-DPA) algorithm [21] DRL using the proximal policy optimization algorithm [22] Mixed-integer programming issue modeled as a stochastic game, utilizing a MARL approach [23] Markov decision process (MDP)-based DRL algorithm using asynchronous advantage actor-critic (A3C) [24] A stochastic policy gradient-based deep neural network (DNN) approach [25]
QoS	Energy efficiency (EE) optimization [26] A nonconvex optimization structure [27] Smart beam-user mapping using Dirichlet distribution [28] A modified version of the simulated annealing (SA) algorithm [29] A dual-layer precoder and cluster-based power allotment scheme [30]	Convolutional neural networks (CNN) [31]
Responsiveness		Comparison of AI-based dynamic algorithms with conventional algorithms [32] Combine conventional optimization and deep learning to reduce computational complexity [33]

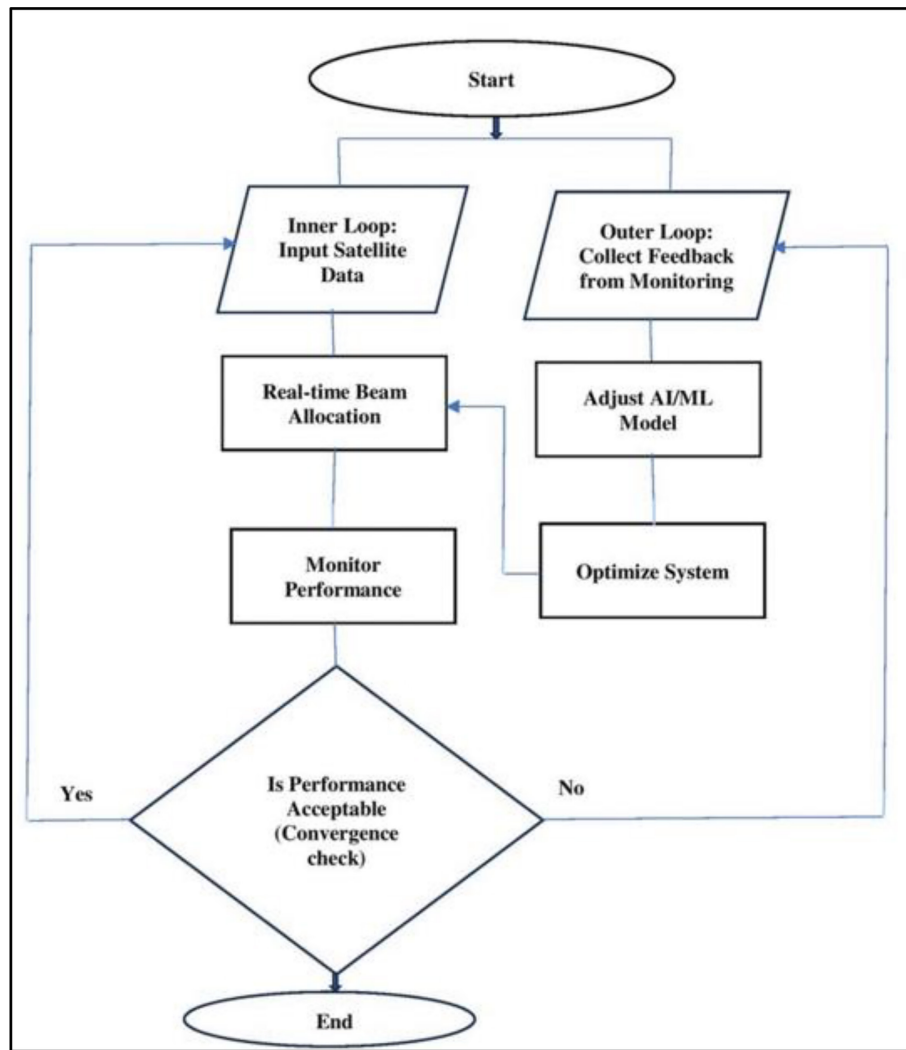


CHART 2 | Resource utilization using DRL-DPA.

payload BW emerges as a crucial consideration for optimizing SatCom systems. Figure exhibits the key challenges in adaptive BW allocation in MBS. This entails dynamically adjusting the bandwidth distribution to accommodate fluctuating demands and ensure optimal utilization of available resources. Table 7 exhibits the key challenges and aspects of dynamic BW allocation affected in MBS. The authors [82] underscore wireless resource management methods tailored for MBS, particularly focusing on the next generation of SatCom networks. Given the anticipation for these networks to uphold exceedingly rapid data rates for smooth integration into forthcoming extensive satellite-ground networks, the primary challenge revolves around diminishing the cost per bit (satellite launch and operation) while enhancing spectral efficiencies. The document emphasizes the necessity for swift and flexible allocation of radio resources based on traffic demand distribution in prospective multibeam broadband satellite systems. It provides an extensive overview of techniques addressing these challenges, encompassing formulation for matching demand, assignment of power and bandwidth, and scheduling design. The research aims to curtail the cost of satellite services, augment flexibility in service provision, and enhance connectivity in underserved areas using nongeostationary orbit (NGSO) satellites. The work summarizes design methodologies, optimization problems, and techniques

for multibeam UHTS systems, addressing both advancements and remaining challenges. Hierarchical optimization strategies, including BW and PA, scheduling, and precoding, are discussed, emphasizing the fundamental optimization principles associated with multibeam SatCom. The letter outlines open challenges and trends in the field, providing valuable insights for future research in multibeam satellite systems. In [80], the focus is on flexible payload operations of SatCom systems, addressing the surging demand for BW in the SatCom industry. With predictions of significant data growth, especially in video and machine-to-machine (M2M) traffic, SatCom plays a pivotal role in extending terrestrial network coverage. This involves connecting maritime, aircraft, and isolated areas, relieving traffic congestion, and enhancing resilience for critical applications. The evolving industry has witnessed the rise of new constellations, ground segment advancements, and innovations in satellite platforms. The paper highlights the need for flexibility in SatCom, especially in GEOs. Flexible payloads, exemplified by recent satellites like quantum, allow for dynamic reconfiguration, impacting payload operations. Traditional missions involve fixed payload requirements, managed over a long lifespan. In contrast, flexible payloads enable dynamic modifications to adapt to changing demands, necessitating automated payload operations. Spare unit management, a fundamental concept in

TABLE 7 | Challenges in BA and aspects affected.

Challenges	Aspect of bandwidth allocation (BA) affected
Beamwidth optimization	Coverage area, frequency reuse, spectrum efficiency
Interference management	Signal quality, frequency planning, resource isolation
Dynamic user demand	Bandwidth allocation flexibility, load balancing
Spectrum efficiency	Optimal use of available bandwidth, minimizing wastage
Resource allocation complexity	Efficient bandwidth distribution, QoS management
Real-time adaptation	Dynamic bandwidth reallocation, response to traffic variability
Power distribution constraints	Coupling of power and bandwidth, maintaining optimal power levels
Coverage area optimization	Beam shaping, effective utilization of bandwidth
Quality of service (QoS) maintenance	Ensuring consistent service, prioritizing critical traffic
Signal degradation	Bandwidth efficiency, interference mitigation

fixed missions, evolves in flexible payloads, eliminating designated spare equipment. The discussion emphasizes the transformation in payload operation concepts, introducing GMV smart payload suite as a solution. This suite addresses the challenges posed by evolving payload architectures and configurations, offering a user-friendly, dynamic, and automated approach to satellite payload operations. The changing landscape prompts a reevaluation of SATCOM payload operation concepts, calling for more efficient, automated, and user-friendly solutions in the evolving SatCom market.

Traditional techniques for adaptive bandwidth allocation in MBS typically involve fixed or static allocation strategies. These methods include fixed bandwidth allocation, where bandwidth is evenly distributed among beams regardless of demand variations, and static bandwidth allocation, where allocation is based on predicted traffic patterns without real-time adjustments. Another approach is using predefined bandwidth plans, where bandwidth is allocated according to predefined templates or historical data. Although straightforward, these conventional techniques lack the flexibility to efficiently manage bandwidth in response to dynamic changes in traffic and user demand. Figure 7 exhibits traditional/conventional techniques used in BW allocation in MBS.

ML and AI techniques in MBS for adaptive bandwidth allocation enable dynamic and data-driven management of bandwidth.

These techniques use algorithms such as deep learning and reinforcement learning to predict traffic demand, optimize bandwidth distribution across beams, and adjust allocations in real-time. By continuously analyzing network conditions and user behavior, ML/AI methods improve efficiency, ensure better QoS, and adapt to changing traffic patterns, leading to more effective and responsive bandwidth utilization in satellite communication networks. Figure 8 exhibits ML/AI techniques used in BW allocation in MBS.

Table 8 provides snapshot of the system performance of both conventional/traditional techniques, and ML/AI techniques in BW allocation. Some techniques are of overlapping in nature as they lie simultaneously in resource utilization, QOS, and responsiveness; hence, the responsiveness attribute is blank.

Chart 3 exhibits the resource utilization in bandwidth allocation using MODRL-TF algorithm, a multiobjective DRL-based approach for two-dimensional time/frequency resource allocation [56]. It can be observed that in the field of BW allocation in MBS, most of the work has been done using traditional or conventional techniques, compared to the relatively limited use of ML and AI techniques. This presents an opportunity to explore and apply ML and AI methods to enhance system performance in adaptive bandwidth allocation.

2.4 | Adaptive Payload BWA

Adaptive Payload BWA for multibeam HTS involves dynamically adjusting the beamwidth of satellite payloads to optimize communication performance. By allocating resources based on changing demand and traffic patterns, this approach enhances spectral efficiency and overall system capacity, ensuring efficient utilization of satellite resources while maintaining high-quality service for users across different regions and applications. The primary objective of beamwidth allocation is to enhance performance metrics. Examples include mitigating the gap between provided and needed capacity or minimizing inaccuracies in coverage. To do this, we need to define factors like service area, satellite position, and beam power. Once we understand the traffic demand across different areas, we divide the service area into sections and assign appropriate beamwidths to match the demand. Areas with higher demand get narrower beams, whereas areas with lower demand get wider beams. The coverage is planned based on these beam configurations and technological limitations. We additionally assess variables such as beam quantity, capacity per beam, total satellite capacity, and cost considerations to ascertain the optimal configuration. Table 9 exhibits potential issues and aspects affected in beamwidth allocation in MBS. A traffic simulator for multibeam SatCom systems, aimed at optimizing resource utilization by understanding spatial-temporal traffic patterns is introduced in [87]. By processing datasets on population distribution, aeronautical, and maritime services, it facilitates the allocation of optimal beamwidths and beam directions to efficiently accommodate varying demand. The simulator aids in modelling satellite traffic patterns, crucial for system design and resource management, with plans to extend its coverage globally in the future. In [88], an optimizing irregular beam coverage and patterns

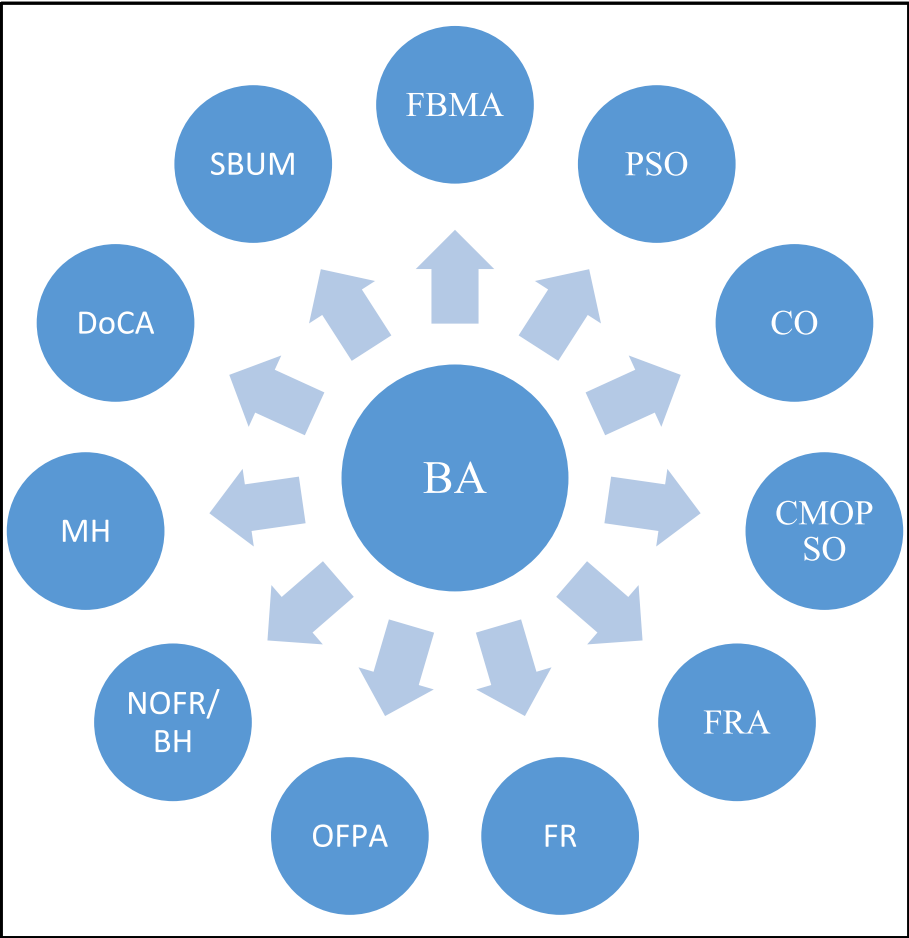


FIGURE 7 | Non-ML techniques in BA.

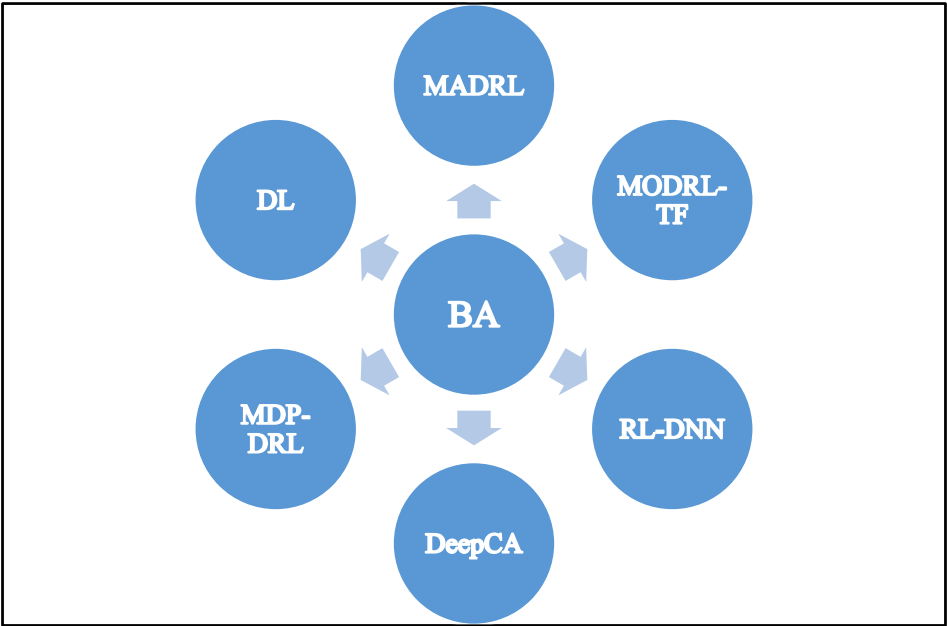


FIGURE 8 | ML-Driven techniques in BA.

in VHTS systems to match nonuniform traffic demands with the objective of lowering costs per gigabit per second in space while minimizing coverage and capacity errors is presented.

By considering technical constraints and comparing them with uniform coverage methods, the methodology offers viable solutions for flexible payload design without significantly

TABLE 8 | System performance of traditional and ML/AI techniques in PA.

System performance	Traditional techniques used	AI/ML techniques used
Resource utilization	Multiobjective optimization, utilizing solutions to enhance efficiency via a Pareto front [12] An analytical method to estimate forward and return capacity, focusing on elements such as effective isotropic radiated power (EIRP), gain over noise temperature (G/T) cumulative densities, carrier-to-interference (C/I) estimations, and capacity assessments for both links [83] A novel SINR formulation demonstrates the frequency/time duality of NOFR and BH schemes [34] A max-min balancing SINR with an analytical solution for optimal carrier allocation [11] A frequency RA method that accounts for interbeam interference using a digital channelizer and extend the conventional flexibility analysis model for more accurate quantification [47] Maximization of the satellite spectrum utilization by pushing the spectrum reuse to affordable limits in terms of tolerable interference [27] Exploring intelligent beam-user assignment with adaptable bandwidth allocation per beam, enhancing payload performance with dynamic radio resource allotment [28] A bandwidth adjustment technique for each beam to reduce the disparity between traffic demand and allocated capacity, ensuring flexible allocation and fairness among spot beams [48] BW allocation as convex optimization, balancing system total capacity and spot beam fairness, using a subgradient algorithm to find optimal bandwidth allocation per beam [84] Simulated annealing algorithm with comprehensive complexity analysis [29] A genetic algorithm combined with repair functions to ensure solution validity and accelerate convergence [13] A metaheuristic algorithm derived from particle swarm optimization, formulated as a multifaceted approach considering system constraints and evaluated using a full-RF link budget model [49] The CMOPSO algorithm to balance convergence speed and diversity across multiple Pareto fronts [50] An AMPL-based frequency assignment algorithm optimizing dynamic plans considering system constraints like handovers and interference [53]	DeepCA, a deep reinforcement learning-based approach for energy-efficient channel allocation in SIoT [55] The MODRL-TF algorithm, a multiobjective deep reinforcement learning-based approach for adaptive, two-dimensional time-frequency resource allocation [56] A DRL-based adaptive beam pattern and bandwidth distribution scheme using a cooperative multiagent framework [44] An AI-driven multiobjective radio resource allocation algorithm using neural network [57]

(Continues)

TABLE 8 | (Continued)

System performance	Traditional techniques used	AI/ML techniques used
QoS	An algorithm for using a physical-mathematical model for rain attenuation prediction is proposed [85] An overview of precoding techniques in current satellite systems, including theoretical development in diverse scenarios, and outlines future pathways for flexible satellite systems [51] High frequency reuse in multiple beams and onboard processing advantages over ground-based methods, with implications for future adaptation in mobile satellite services and mega constellations [86] Utilizing flexible MBS and user distribution for enhanced satellite-assisted edge caching systems, optimizing BW allocation and caching decisions to minimize cache feeding time and maximize cache hits, overcoming nonconvexity with a proposed iterative DC algorithm [52]	A novel DRL-based DCA algorithm is proposed to minimize service blocking probability [54] Combination conventional optimization and DL to lower computational complexity [33]
Responsiveness		

increasing costs. It sets a foundation for future optimization techniques like GAs or ML to further refine satellite payload design. Improvement in HTS systems by adapting beam patterns to accommodate dynamic user demand in remote areas like airplanes and ships is explored in [68]. It critiques fixed multibeam patterns for their inability to handle nonuniform demand and proposes an adaptive approach using beamforming capabilities. Dynamic multibeam configuration and coverage area strategy, incorporating features like elliptical beams and weighted clustering of users, aims to evenly distribute system demand across beams. Simulations indicate superior performance compared to traditional fixed plans, ensuring better coverage and optimized system capacity utilization.

Traditional techniques for adaptive beamwidth allocation in MBS often involve fixed beamwidth configurations that do not adapt to changing conditions. Common approaches include fixed beamwidths, where each beam's width is predetermined and remains constant, and static allocation, where beamwidths are set based on anticipated coverage needs without the ability to adjust in real time. These conventional methods are simple to implement but lack the flexibility to optimize coverage and capacity dynamically, leading to inefficiencies in scenarios with varying user distributions and traffic demands. Figure 9 exhibits its traditional/conventional techniques used in beamwidth allocation in MBS.

ML and AI techniques in MBS for adaptive beamwidth allocation enable dynamic and intelligent adjustment of beamwidths based on real-time data. These techniques use algorithms such as reinforcement learning and optimization models to analyze user distribution, traffic patterns, and environmental factors, allowing beams to be widened or narrowed as needed. By continuously learning from network conditions, ML/AI methods optimize coverage, enhance capacity, and improve overall system efficiency, making beamwidth allocation more responsive to changing demands and ensuring better utilization of satellite resources. Figure 10 exhibits traditional/conventional techniques used in beamwidth allocation in MBS. Table 10 exhibits

the system performance of conventional/traditional techniques and ML/AI used in BWA in MBS.

Chart 4 exhibits the resource utilization in beamwidth allocation using weighted k-means clustering an unsupervised ML technique.

It can be observed that in the field of beamwidth allocation in MBS, most of the work has been done using traditional or conventional techniques, with relatively limited application of ML and AI methods. This presents an opportunity to explore and apply ML and AI techniques to enhance system performance in beamwidth allocation, which is directly related to the gain of the satellite antenna.

Based on the review of literature on resource allocation in terms of BH, PA, BA, and BWA, a survey framework is developed based on attributes such as resource utilization, power management, QoS, and responsiveness for both traditional and AI/ML-based techniques. Table 11 exhibits the survey framework for both techniques.

3 | Opportunities, Challenges, and Evaluation of Techniques in MBS

3.1 | Opportunities and Challenges in MBS

MBS systems offer significant advancements in communication technology, enabling enhanced coverage, capacity, and flexibility. However, they also come with their set of challenges that need to be addressed for optimal performance. Although the paper primarily focuses on computational complexity, it is indeed crucial to consider how these techniques align with and impact broader industry standards and practices, such as those developed by 3GPP and DVB. Here's why:

1. Alignment with standards: Understanding how resource allocation techniques, including both conventional methods

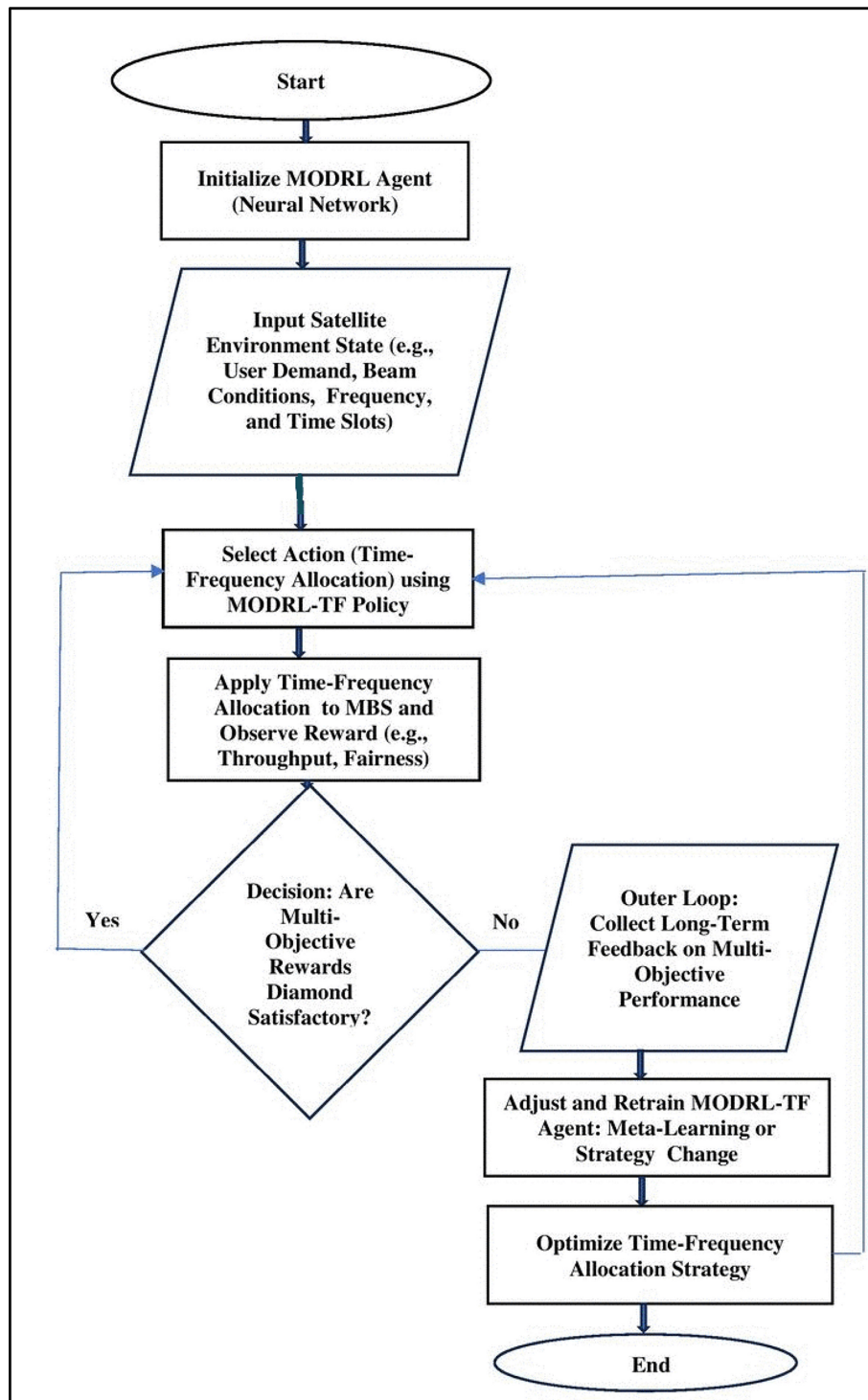


CHART 3 | Resource utilization using MODRL-TF

and AI/ML approaches, fit within international standards is essential for ensuring that these technologies can be effectively implemented and integrated into existing SatCom and NTN infrastructures. Standards set by organizations like 3GPP and DVB provide guidelines for interoperability, performance, and compatibility, which are critical for the successful deployment of these technologies.

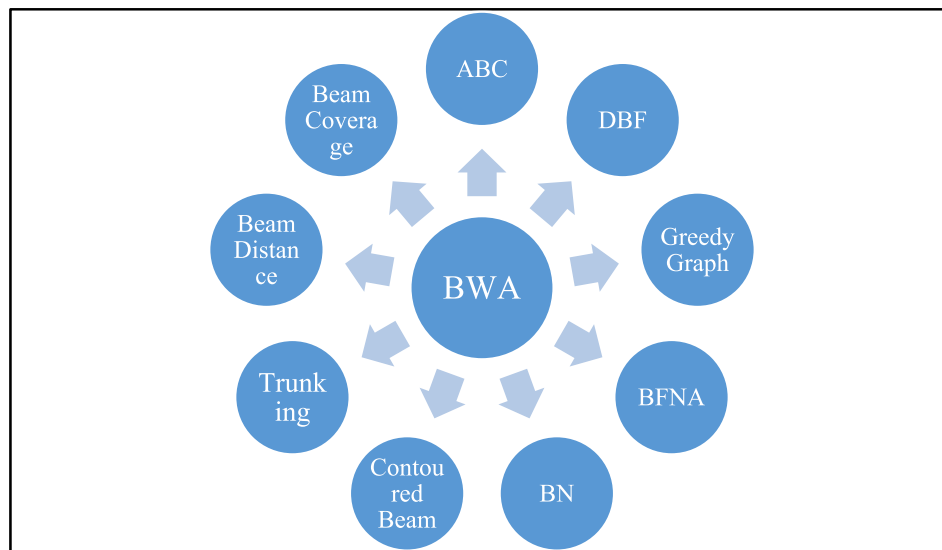
2. Industry relevance: The technologies discussed in the paper are indeed relevant to current and future industry developments. By addressing how these techniques are

being considered in standardization efforts, the paper can provide a more comprehensive view of their practical applications and potential challenges. This will not only enhance the paper's relevance to industry stakeholders but also offer a clearer perspective on how these techniques can be adopted in real-world scenarios.

3. Recommendation for inclusion: To address this, I strongly suggest incorporating a discussion on how resource allocation techniques are being evaluated and developed in the context of international standards. This could include:

TABLE 9 | Challenges in BWA and aspects affected.

Challenges	Beamwidth allocation strategies
Beam coverage optimization	Efficient allocation of beamwidth to ensure optimal coverage for all users
Dynamic user demand	Adjusting beamwidth allocation in response to fluctuating user demand
Interference management	Mitigating interference between beams to ensure effective bandwidth allocation
Spectrum efficiency	Maximizing the use of available spectrum through optimized beamwidth distribution
Resource allocation complexity	Managing the complexity of distributing beamwidth resources across multiple beams
Real-time adaptation	Adjusting beamwidth allocation in real-time to adapt to changing conditions
Power distribution constraints	Allocating beamwidth in conjunction with limited power resources for optimal performance
Quality of service (QoS) maintenance	Ensuring that beamwidth allocation supports consistent QoS for all users
Signal degradation	Preventing signal degradation by optimizing beamwidth allocation

**FIGURE 9** | Non-ML techniques in BWA.

- Current standardization efforts: A brief overview of how techniques like beam hopping, power allocation, and bandwidth management are being standardized by 3GPP, DVB, or other relevant bodies.
- Impact on implementation: An analysis of how compliance with these standards affects the implementation and performance of AI/ML approaches compared to conventional methods.
- Applications: Forward error correction (FEC) decoding algorithms, such as LDPC, Reed–Solomon, or polar codes, are essential for modern digital communications based on 3GPP and DVB standards. These algorithms demand high complexity and power, motivating AI acceleration through ML algorithms to reduce these requirements for satellite payloads. Conventional FEC relies on maximum a posteriori decoding, with ML application restricted to shortcodes due to exponential training complexity and

challenges like unknown channel noise distribution [90]. The belief propagation (BP) algorithm, trained as a NN, is the most popular ML-based approach to enhance error correction performance. Reducing FEC decoding complexity for power-limited regenerative satellites, such as those using DVB-S2X, enables scalability to ultra-high-throughput satellites (UHTSs) with thousands of data-links. ML-based approaches can address challenges like high computational complexity and channel noise uncertainty, particularly for shortcodes. Two ML techniques, artificial neural networks (ANNs) and recurrent neural networks (RNNs), are considered for onboard FEC decoding. Whereas ANNs have fixed input sizes, RNNs can process sequences of arbitrary length, making them suitable for systems with varying conditions, like cognitive radio [91]. RNNs excel at error correction by leveraging context from supervised training, making them a preferred choice for regenerative satellites. In [92], a

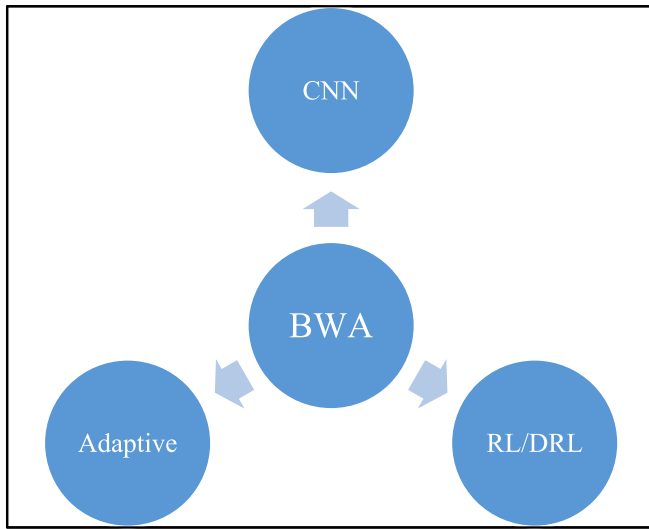


FIGURE 10 | ML-driven techniques in BWA.

decoding approach is presented for 5G data frames using autoencoders and deep NNs, outperforming traditional LDPC-based methods by achieving a 3 dB SNR gain in AWGN channels. This method, potentially applicable to satellite integration in 5G networks, could further benefit from MIMO techniques for spatial diversity. In the area of integrated satellite-terrestrial networks, AI-based solutions are proposed in [93] to manage network slices in integrated 5G-satellite networks, addressing challenges like predictable and unpredictable network variations. Predictable variations are tackled using time-expanded graphs, whereas unpredictable changes, such as mega-constellation topology shifts, require rapid adaptation to model drift. Proposed solutions include conventional learning, transfer learning, meta-learning, and reinforcement learning. Integrating network slicing with ML to optimize links in 6G-satellite networks is presented in [94], demonstrating a ranking-based framework that improves user acceptance ratios compared to traditional methods.

TABLE 10 | System performance of traditional and ML/AI techniques in PA.

System performance	Traditional techniques used	AI/ML techniques used
Resource utilization	Optimization of gain of hexagonally arranged uniform beams by addressing a 4 dB variation at beam crossovers while balancing low sidelobe levels, beam count, power, and traffic flexibility, and size and mass constraints [58] High frequency reuse for efficient use of limited frequency resources [60] Use of progressive pitch and coverage expanding methods, alongside complex global resource management [62] A novel mathematical model to resolve the transmit power and beam directivity trade-off [63] Optimizing resources, processing, and architectures in next-generation LEO systems [65] Establishing relationships between intraband and interband spot beam distances and total system throughput [66]	CNN-based resource management in flexible payload architectures [31] An adaptive multibeam pattern with flexible spot beam size and positioning to enhance satellite system flexibility [68]
QoS	Exploring design considerations for reflector antennas and feed assemblies, with emphasis on coverage gain, sidelobe isolation, and cross-polar isolation [59] A randomized multistart heuristic using graph recoloring, local search, and SA to develop a nonuniform layout [61] A beam control technique for remote sensing satellites in Q band to optimize signal-to-noise power ratio at ground stations while minimizing interference with terrestrial networks [64] A new satellite multicast scheme using phased array antennas with flexible beam direction and width, partitioning users into groups allocated to appropriate beams to optimize average data rate and ensure QoS, supported by an iterative algorithm [89]	RL and DRL algorithms—Q-learning (QL), deep Q-learning (DQL), and double deep Q-learning (DDQL)—compared for performance, complexity, and latency [67]
Responsiveness		

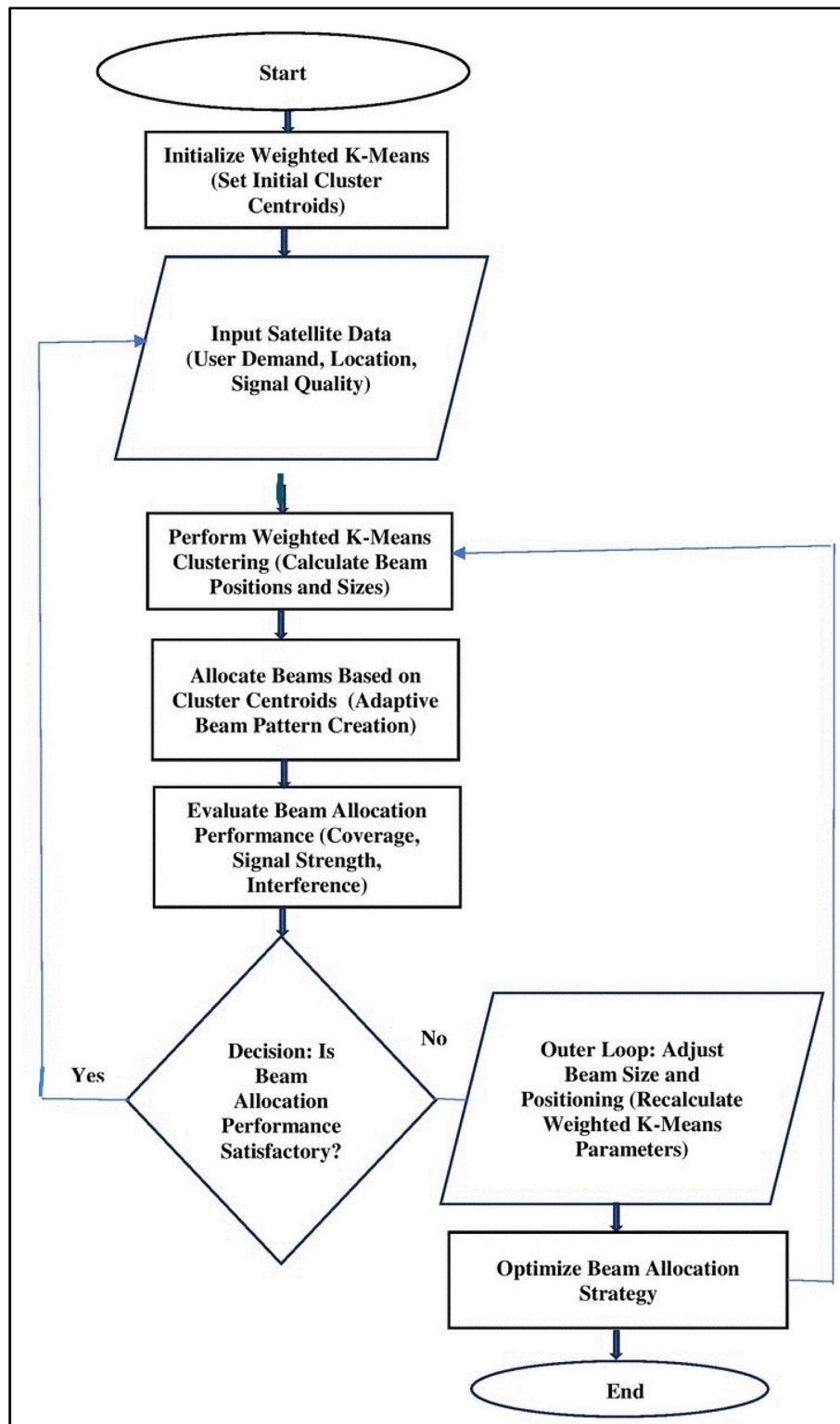


CHART 4 | Resource utilization using weighted k-means.

Similarly, a solution combining queuing theory and NNs is developed in [95] to minimize latency in end-to-end 5G-satellite communications.

- Future directions: AI and ML techniques play a pivotal role in the evolution of the industrial standardization processes of 3GPP and DVB to accommodate the demands of

next-generation networks and services. Both 3GPP and DVB have dedicated working groups for incorporating AI/ML. Whereas 3GPP focusses on AI-based service architecture, media handling, and new use cases, DVB explores the application of AI in broadcasting workflows. AI/ML addresses several key standardization challenges such as interoperability, latency reduction, security, and

TABLE 11 | Survey framework for traditional and AI/ML techniques.

Attributes	Traditional techniques	AI/ML techniques
Resource utilization	Predictable and relatively low computational requirements Modest data requirements and focused on specific variables of interest Less flexible and adaptive; rule changes and updates require manual intervention	Often require significant computational power, especially during training phases Depend heavily on large volumes of high-quality data for training and validation Highly adaptive and can automatically improve with new data
Power management	Less computational power requirements Predictable and stable energy requirements once the system is in operation Often run on older, less energy-efficient hardware Systems often run at a consistent power level with little need for dynamic adjustments	Require significant computational resources, particularly during the training phase Energy consumption can vary widely depending on the complexity of the model, the size of the dataset, and the phase (training vs. inference) Utilizes cutting-edge hardware optimized for AI tasks and consumes significant power Utilizes dynamic power management strategies, which adjusts computational resources based on demand
QoS	Predictable response time Low latency Deterministic accuracy Limited adaptability High reliability Ease of debugging Limited scalability	Variable response time Potential higher latency High accuracy for complex tasks Adaptive learning Variable reliability Potential for unpredictable errors High scalability
Responsiveness	Low latency Deterministic execution Static response Manual updates required Predictable scalability Resource intensive Limited to predefined scenario Deterministic but rigid Stable but slow to adapt	Variable latency Slower response Dynamic response Self-improvement possible Efficient scaling Elastic scaling. High complexity handling Adaptable and flexible Continuous learning and proactive adjustment possible

scalability. Various stages of 3GPP standardization for cellular networks, including LTE, 5G, and the upcoming 6G, employ AI-driven solutions and ML techniques for network optimization and resource management [96, 97]. AI algorithms are used for predicting the traffic patterns and optimizing spectrum usage across diverse scenarios, detection of anomalies, and predict failures, reducing network downtime. ML is used to automate parameter adjustments to maximize network efficiency. Because DVB focuses on standards for digital television and broadcasting, AI/ML is instrumental in enhancing service delivery and user experience by optimization of content delivery, personalization, and user experience, enhancing the signal processing and managing the resource in hybrid systems.

Table 12 provides a snapshot of the opportunities, challenges, and computational complexities involved in using both methods.

Opportunities:

Enhanced coverage and capacity:

- MBS can provide broader coverage and increased capacity compared to traditional single-beam satellites. By employing multiple beams, these systems can serve a larger number of users simultaneously across different geographic regions.

Flexibility and adaptability:

- Flexibility and adaptability are key attributes of MBS and are essential for optimizing their performance and ensuring their ability to meet diverse and changing demands to adapt to changing conditions. These attributes are critical for optimizing satellite performance, enhancing service quality, and ensuring the efficient use of resources. Beam-hopping technology allows satellites to dynamically switch

TABLE 12 | Opportunities and challenges.

Aspect	Opportunities	Challenges	Computational Complexity	
			Non-ML	ML
Beam hopping	Enhanced adaptability and resource usage	Coordination complexity and QoS maintenance	Rely on traditional optimization techniques and algorithms Possess lower initial computational complexity but can become computationally expensive and less adaptable as the problem size increases and dynamic conditions change	Use data-driven methods to optimize beam hopping patterns Possess higher initial computational complexity due to the training process but offer potentially lower complexity and better performance in real-time operation, adapting more efficiently to changing patterns and environments
Power allocation	Increased system capacity and efficiency	Optimization amidst regulatory constraints		
Bandwidth allocation	Improved spectral efficiency	Balancing varying demands and congestion		
Beamwidth allocation	Enhanced coverage and capacity	Optimizing for user distribution and interference		

between beams based on changing demand patterns or environmental conditions. This flexibility enhances system adaptability and ensures efficient resource utilization. Additionally, use of advanced onboard processing, adaptive interference mitigation, and other innovative techniques can provide high-quality, reliable, and efficient services across a wide range of scenarios. Existing satellite operations can be enhanced substantially with the use of these flexible and adaptive techniques, thus evolving with technological advancements and emerging market needs.

Improved spectral efficiency:

- Efficient power, BW, and BWA techniques can lead to improved spectral efficiency in multi-beam satellite systems. By optimizing RA, these systems can maximize data throughput while minimizing interference.

Challenges:

Beam-hopping coordination:

- Coordinating beam-hopping across multiple beams introduces complexities in satellite system design and operation. Ensuring smooth transition between beams while upholding QoS for users poses a notable challenge.

Power allocation optimization:

- Optimizing PA among different beams is essential to maximize system capacity and minimize interference. However, achieving efficient power control while adhering to regulatory constraints and minimizing power consumption poses technical challenges.

BA management:

- Efficiently allocating BW among multiple beams requires sophisticated resource management algorithms. Balancing the varying BW demands of different users and applications while avoiding congestion and ensuring fairness is a complex task.

BWA for coverage optimization:

- Allocating beamwidth appropriately to different beams is crucial for optimizing coverage and capacity. However, determining the optimal beamwidth configuration while considering factors such as user distribution, terrain variations, and interference mitigation presents a significant engineering challenge.

3.2 | Comparison of Conventional and AI/ML Methods

Conventional methods:

- Conventional beamforming: Conventional beamforming techniques involve static beamforming patterns that are predetermined based on system design parameters. These patterns are often manually optimized and may not adapt

TABLE 13 | Comparison of both methods.

Aspect	Conventional methods	AI/ML-driven approaches
Beamforming	Static beamforming patterns	Adaptive beamforming with RL algorithms
Resource allocation	Rule-based allocation schemes	ML for DRA
Complexity	Often simpler, but less adaptive	More complex, but adaptable to dynamic changes
Adaptability	Limited adaptability to changing conditions	Enhanced adaptability and optimization
Optimization	Manual optimization with predefined rules	Data-driven optimization based on real-time feedback

well to dynamic changes in user distribution or environmental conditions.

- Rule-based RA: Traditional RA schemes rely on predefined rules and heuristics for beam, power, BW, and BWA. Although these methods may offer simplicity, they lack adaptability and optimization capabilities in complex and dynamic satellite environments.

AI/ML-driven approaches:

- Adaptive beamforming with reinforcement learning: AI/ML-driven adaptive beamforming techniques leverage reinforcement learning algorithms to dynamically adjust beamforming patterns based on feedback from the satellite environment. This enables improved adaptation to changing user demands and environmental factors.
- ML for DRA: AI/ML algorithms possess the capability to scrutinize live data and patterns of user behavior in real time, enabling the dynamic optimization of RA. Through the assimilation of historical data and the adjustment to evolving circumstances, such methodologies can elevate the efficiency and performance of systems.

Table 13 exhibits comparison of conventional methods and AI/ML methods.

The satellite industry can harness the full potential of AI/ML technologies to enhance the performance, reliability, and adaptability of MBS systems, ultimately driving innovation and improving global connectivity. Initially, static models and fixed algorithms, for example, classical signal processing techniques, were employed in beamforming and beam steering and to tackle electromagnetic interference. For FR and maximizing the throughput, traditional techniques such as linear programming and heuristic algorithms were used. For traffic management, statistical techniques were used. The AI/ML models facilitate adaptive and intelligent distribution of resources based on real-time data, which enhance the efficiency in allocation of resources. Several techniques such as deep learning, predictive analytics, reinforcement learning, and neural networks have been employed for this purpose.

However, due to the complexity and hard decision-making processes, many AI/ML models, particularly deep learning networks, are often considered as “black boxes.” Furthermore, the development and refinement of AI/ML models is largely limited

due to nonaccessibility to comprehensive MBS datasets, and a big challenge is encountered in ensuring compatibility and smooth integration of these techniques in existing infrastructure.

The integration of traditional and AI/ML techniques offers a balanced approach that harnesses the strengths of both paradigms and provides a balanced, robust, and efficient approach to problem-solving. This hybrid approach not only enhances accuracy, reliability, and interpretability but also ensures scalability, adaptability, and cost-effectiveness. The complementary strengths of both methodologies offer superior outcomes, foster innovation, and enable the development of cutting-edge solutions in an increasingly data-driven world.

4 | Conclusion

MBS are a vital component of modern communication infrastructure, providing high-speed Internet, broadcasting, and other telecommunications services to a wide area of the globe. Its adaptive characteristics can attain efficient and reliable communication services across diverse geographical regions under changing environmental and operating conditions. In the present work, an exhaustive review is carried out in terms of four important aspects of MBS, that is, payload beam hopping, along with allocations of power, BW, and beamwidth focusing on both non-ML and ML-driven techniques separately. It is suggested that a combination of non-ML and ML-driven techniques in MBS can offer higher performance, increased spectral efficiency, improved reliability, and better adaptability to dynamic and unpredictable communication environments to further optimize the performance and enhance capabilities of MBSSs.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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